ReLiSS: Modality-Adaptive Representation Conditioning for Multimodal MRI Segmentation under Arbitrary Modality Availability

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***Abstract*— Multimodal magnetic resonance imaging (MRI) supports brain-lesion segmentation, but missing sequences and variation among available inputs complicate multimodal integration. We present ReLiSS, a unified framework for 3D segmentation with arbitrary modality subsets, enabling adaptive feature conditioning and uncertainty estimation under incomplete inputs. Availability-Gated Spectral Conditioning (AGSC) derives modality scores from compact spatial–spectral descriptors and suppresses absent inputs through deterministic availability gating. These scores guide input attenuation, adaptive fusion, and multi-scale feature-wise linear modulation within a hierarchical large-kernel CNN–Mamba encoder and UNETR-style decoder. Task-Matched Evidential Learning (TM-EDL) derives Beta or Dirichlet evidential vacuity from segmentation logits, while Nested-View Evidential Ranking (NV-Rank) encourages vacuity ordering between parent and child modality subsets. Both components regularize training without additional learned inference-time branches. Evaluation on BraTS2021, ISLES2022, WMH2017, and a private tuberous sclerosis complex cohort covers two- to four-modality settings and 28 incomplete-modality configurations. ReLiSS achieves the highest mean Dice among the evaluated methods under incomplete inputs (80.82%, 65.08%, 70.68%, and 64.20%, respectively), while maintaining competitive full-modality performance. Ablation studies support the complementary contributions of input-adaptive conditioning and nested-view evidential regularization. These findings support their combined use for segmentation with incomplete multimodal MRI.**

***Index Terms*— Brain lesion segmentation, incomplete modalities, multimodal magnetic resonance imaging, modality-adaptive conditioning, state-space models, evidential learning.**

Code Availability: Source code and pretrained model weights are publicly available at `https://github.com/hekaiwang/ReLiSS`.

I. INTRODUCTION

Multimodal magnetic resonance imaging (MRI) provides complementary anatomical and pathological information for brain-lesion segmentation, although the contribution of each sequence varies across targets and regions [1], [2]. In practice, heterogeneous protocols, motion artifacts, scanner constraints, contrast-agent contraindications, and incomplete archives can leave required modalities unavailable, compromising segmentation performance.

Existing approaches address missing modalities through modality completion [3], [4], knowledge distillation [5], [6], or unified multimodal learning [7]–[10]. Recent studies explore masked or contrastive representation learning, state-space modeling, modality-impact calibration, and imbalanced modality availability during training [2], [11]–[15]. These approaches vary in their use of reconstruction branches, teacher–student pipelines, modality-specific encoders, and availability indicators. Binary availability identifies which modalities are present but does not characterize variation among observed inputs. Standard segmentation supervision also leaves the behavior of confidence-related signals across nested modality subsets unconstrained.

ReLiSS addresses these issues through input-adaptive representation conditioning and nested-view evidential regularization. Availability-Gated Spectral Conditioning (AGSC) derives modality scores from compact spatial–spectral descriptors of zero-filled inputs and suppresses missing modalities with a deterministic availability gate. The resulting score vector guides input attenuation, adaptive fusion, and multi-scale feature-wise linear modulation (FiLM) within a large-kernel CNN–Mamba encoder and UNETR-style decoder. Task-Matched Evidential Learning (TM-EDL) derives evidential vacuity from segmentation logits, and Nested-View Evidential Ranking (NV-Rank) regularizes vacuity across nested modality subsets.

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Both evidential components operate only during training and add no learned inference-time branch.

The main contributions are summarized as follows:

1) Availability-Gated Spectral Conditioning combines spatial–spectral input cues with deterministic availability, enabling input-dependent modality weighting while guaranteeing zero contribution from missing inputs.
2) Score-guided fusion and multi-scale FiLM conditioning are integrated into a unified CNN–Mamba encoder–decoder that handles arbitrary nonempty modality subsets without modality reconstruction, externally supplied availability vectors, or subset-specific models.
3) TM-EDL and NV-Rank jointly regularize task-matched evidential vacuity across nested modality views during training. Evaluation on four datasets covers 28 incomplete-modality configurations across two-, three-, and four-modality settings.

II. RELATED WORK

A. Learning and Fusion with Missing Modalities

Missing-modality segmentation has been addressed through hetero-modal aggregation, modality completion, representation decomposition, knowledge transfer, and adaptive fusion. Representative approaches include HeMIS and U-HVED [3], [7], shared–specific or disentangled representation learning [16]–[19], and knowledge-transfer methods such as ACN, SMU-Net, and AdaMM [5], [6], [20].

Adaptive fusion uses region-aware or attention-based interactions in RFNet, mmFormer, and M^{2}FTrans [8]–[10], task- or consistency-guided weighting in SimMLM and CLoE [21], [22], and spectral cues in frequency-aware methods [23], [24]. PNDC calibrates region-level fusion to mitigate negative modality impacts [2]. ReLiSS instead derives modality-level spatial–spectral scores to guide input attenuation, adaptive fusion, and multi-scale feature conditioning.

Masked and contrastive learning offers another approach. M^{3}AE uses modality- and patch-level masking, while UniME combines a pretrained unified ViT representation with modality-specific CNN features [4], [14], [25]. CLRS synthesizes missing modality-specific latent representations with cyclic synthesis and contrastive alignment, whereas MCPL learns modality and class prototypes through dual-view contrastive objectives [26], [27]. Reconstruction- and prompt-based adaptation has also been explored [28], including LFC-SATP-SGG, which combines latent feature completion, missing-state text prompting, and semantic-guided gating [29]. HGA addresses imbalanced modality availability in training cohorts by aligning gradient direction and magnitude across modality combinations and unimodal branches [15]. ReLiSS instead conditions a single segmentation model on observed inputs and combines this adaptation with subset-structured evidential regularization, without reconstructing missing images or latent features.

B. Volumetric Architectures and State-Space Modeling

Volumetric segmentation has evolved from convolutional encoder–decoders to large-kernel ConvNets and transformer-based architectures [30]–[32]. Mamba introduces selective state-space modeling [33], which has been extended to 3D segmentation by SegMamba [34]. For incomplete multimodal segmentation, LS3M, IM-Fuse, and M^{2}SegMamba use state-space modeling for token ordering, cross-modal fusion, or masked representation learning [11]–[13]. ReLiSS instead employs Mamba as a deep-stage context module within a modality-adaptive conditioning framework.

C. Evidential and Subset-Structured Regularization

Predictive uncertainty has been studied through Bayesian approximations, deep ensembles, and evidential learning [35]–[37], including multi-view evidential fusion [38]. Reliable uncertainty estimation nevertheless requires dedicated calibration and robustness evaluation [39]–[41].

SIUM uses Gaussian representations and modality-set inclusion to regularize uncertainty across incomplete inputs [42]. ReLiSS adopts a related subset-structured principle with a different representation and training role: Beta or Dirichlet vacuity is derived directly from segmentation logits and aggregated over a shared region of interest (ROI). TM-EDL and NV-Rank use this quantity as an auxiliary training signal without an independent uncertainty head.

III. METHOD

A. Framework Overview

ReLiSS integrates AGSC, a hierarchical CNN–Mamba encoder–decoder, TM-EDL, and NV-Rank (Fig. 1). For each dataset, the input comprises a predefined set of M MRI modalities, with missing channels filled by exact zeros. A single model handles all $2^M - 1$ nonempty subsets; for BraTS, $M = 4$ with T1, T1ce, T2, and FLAIR. AGSC estimates modality scores for input attenuation, adaptive fusion, and multi-scale FiLM conditioning, and the encoder–decoder produces segmentation logits. TM-EDL derives task-matched evidential vacuity from these logits, which NV-Rank regularizes across nested modality views during training. Neither component adds a learned inference-time branch.

B. Availability-Gated Spectral Conditioning

For each zero-filled modality $\widetilde{x}_m$, AGSC extracts a compact six-dimensional spatial–spectral descriptor characterizing spatial intensity, global spectral magnitude, low- and high-frequency composition, spectral dispersion, and their relative balance. Spectral statistics are computed from the magnitude of a one-sided 3D Fourier transform using a normalized radial threshold of 0.25. After modality-specific standardization, the descriptor $\bar{\mathbf{d}}_m$ and modality identity $\mathbf{e}_m$ are mapped by a shared MLP to an input-dependent conditioning score. With $\epsilon_a = 10^{-7}$,

$$\begin{aligned} q_m &= \sigma\big(\phi_{\mathrm{q}}([\bar{\mathbf{d}}_m; \mathbf{e}_m])\big), \\ a_m &= \mathbb{I}[\mathrm{mean}\,|\widetilde{x}_m| > \epsilon_a], \qquad r_m = a_m q_m. \end{aligned} \tag{1}$$

Here, a_m denotes modality availability, q_m is an input-dependent conditioning score, and r_m is its availability-gated counterpart. Missing modalities receive exactly zero scores.

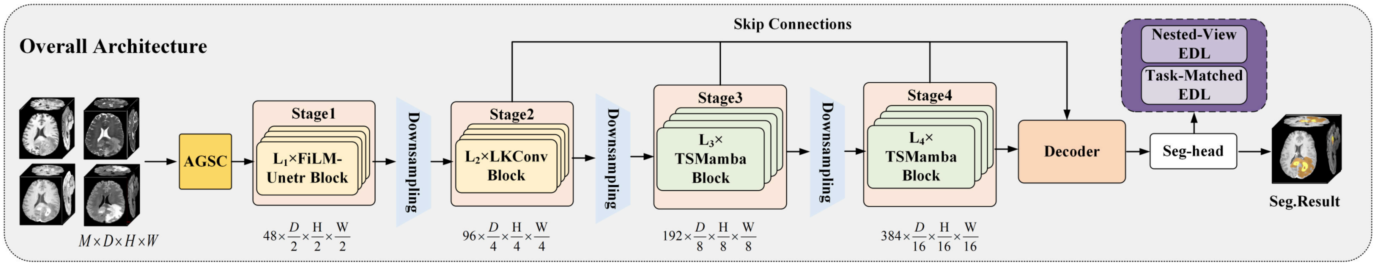

Overall Architecture
Skip Connections
$M\times D\times H\times W$
AGSC
Stage1
L1×FiLM-Unetr Block
$48\times\frac{D}{2}\times\frac{H}{2}\times\frac{W}{2}$
Downsampling
Stage2
L2×LKConv Block
$96\times\frac{D}{4}\times\frac{H}{4}\times\frac{W}{4}$
Downsampling
Stage3
L3× TSMamba Block
$192\times\frac{D}{8}\times\frac{H}{8}\times\frac{W}{8}$
Downsampling
Stage4
L4× TSMamba Block
$384\times\frac{D}{16}\times\frac{H}{16}\times\frac{W}{16}$
Decoder
Nested-View EDL
Task-Matched EDL
Seg-head
Seg.Result

Fig. 1. Overview of ReLiSS. Zero-filled multimodal inputs are first processed by AGSC and then encoded by a four-stage hierarchical CNN–Mamba backbone with multi-scale skip connections. A UNETR-style decoder recovers full-resolution segmentation logits. TM-EDL and NV-Rank regularize the segmentation output during training, while only the main segmentation pathway is retained at inference.

The scoring function is optimized jointly with the segmentation objective without requiring external modality-quality labels or degradation-level supervision.

Modality-specific stems h_m process the score-conditioned inputs, which are adaptively fused as

$$\begin{aligned} w_m &= \frac{a_m \exp(\tau r_m)}{\sum_{j=1}^{M} a_j \exp(\tau r_j)}, \\ \mathbf{F}^{(0)} &= \sum_{m=1}^{M} w_m h_m(r_m \widetilde{x}_m), \end{aligned} \tag{2}$$

where $\tau > 0$ is learnable. Missing modalities receive zero fusion weight, and observed modalities are weighted according to their conditioning scores. The score vector $\mathbf{r} = [r_1, \ldots, r_M]$ also modulates the fused stem and multi-scale encoder features through FiLM. Its final projections are zero-initialized to introduce conditioning progressively during optimization.

C. Hierarchical CNN–Mamba Segmentation Backbone

The segmentation backbone contains four hierarchical stages with channel widths $[48, 96, 192, 384]$. Early stages employ residual large-kernel convolutional blocks to capture local and multi-scale representations, whereas the two deepest stages incorporate TSMamba blocks for long-range spatial modeling. Each TSMamba block scans the feature volume along the depth, height, and width directions with shared parameters and aggregates the restored directional features through a residual connection. Axis-aware downsampling avoids excessive resolution reduction along short spatial dimensions.

A UNETR-style decoder [32] combines bottleneck features with score-conditioned multi-scale skip connections to recover full-resolution segmentation logits. Deep supervision is applied during training. Final segmentation probabilities are obtained directly from the logits using the task-appropriate sigmoid or softmax, while the evidential formulation described below serves only as an auxiliary training regularizer.

D. Task-Matched Evidential Learning

TM-EDL derives task-compatible evidential vacuity from segmentation logits for auxiliary regularization. Define

$$\mathcal{E}(z) = \operatorname{clip}(\operatorname{softplus}(z), 0, e_{\max}), \qquad e_{\max} = 80. \tag{3}$$

For a mutually exclusive K-class task,

$$\begin{aligned} &\widetilde{z}_k(v) = z_k(v) - \frac{1}{K}\sum_{j=1}^{K} z_j(v), \\ &\alpha_k(v) = \mathcal{E}(\widetilde{z}_k(v)) + 1, \qquad u(v) = \frac{K}{\sum_{k=1}^{K}\alpha_k(v)}. \end{aligned} \tag{4}$$

Centering makes vacuity invariant to a common logit shift. For an overlapping binary region r,

$$\alpha_r^{\pm}(v) = \mathcal{E}(\pm z_r(v)) + 1, \qquad u_r(v) = \frac{2}{\alpha_r^{+}(v) + \alpha_r^{-}(v)}. \tag{5}$$

BraTS uses separate Beta parameterizations for WT, TC, and ET. The resulting vacuity is derived directly from the segmentation logits and is used as a task-matched auxiliary signal for evidential regularization.

E. Nested-View Evidential Ranking

NV-Rank encourages vacuity ordering across nested modality views. For each eligible sample b, parent and child sets $\mathcal{S}_p$ and $\mathcal{S}_c = \mathcal{S}_p \setminus \{m_b\}$, with $|\mathcal{S}_p| \geq 2$, are processed by the same network using identical augmentation and a shared ROI. Let $U^{p}_{b,h}$ and $U^{c}_{b,h}$ denote ROI-averaged vacuity for output head h. We define

$$\omega_b = \operatorname{sg}\left(\frac{r_{b,m_b}}{\max_{j\in\mathcal{S}_p} r_{b,j} + \epsilon}\right), \tag{6}$$

$$\mathcal{L}_{\text{mono}} = \frac{\sum_{(b,h)\in\mathcal{I}} \omega_b [U^{p}_{b,h} - U^{c}_{b,h}]_+}{\sum_{(b,h)\in\mathcal{I}} \omega_b + \epsilon}, \tag{7}$$

where sg denotes stop-gradient, $\mathcal{I}$ contains valid sample–head pairs, $[x]_+ = \max(x, 0)$, and ϵ ensures numerical stability. The ranking objective encourages consistent evidential ordering between nested modality views over the shared ROI, thereby introducing subset-structured regularization during training.

F. Training Objective

All trainable components of ReLiSS, including AGSC and the segmentation network, are optimized jointly from scratch with modality dropout and standard data augmentation. No auxiliary modality-score labels or degradation-level supervision are required. The overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + 0.10\mathcal{L}_{\text{EDL}} + 0.02\chi_t\mathcal{L}_{\text{mono}}, \tag{8}$$

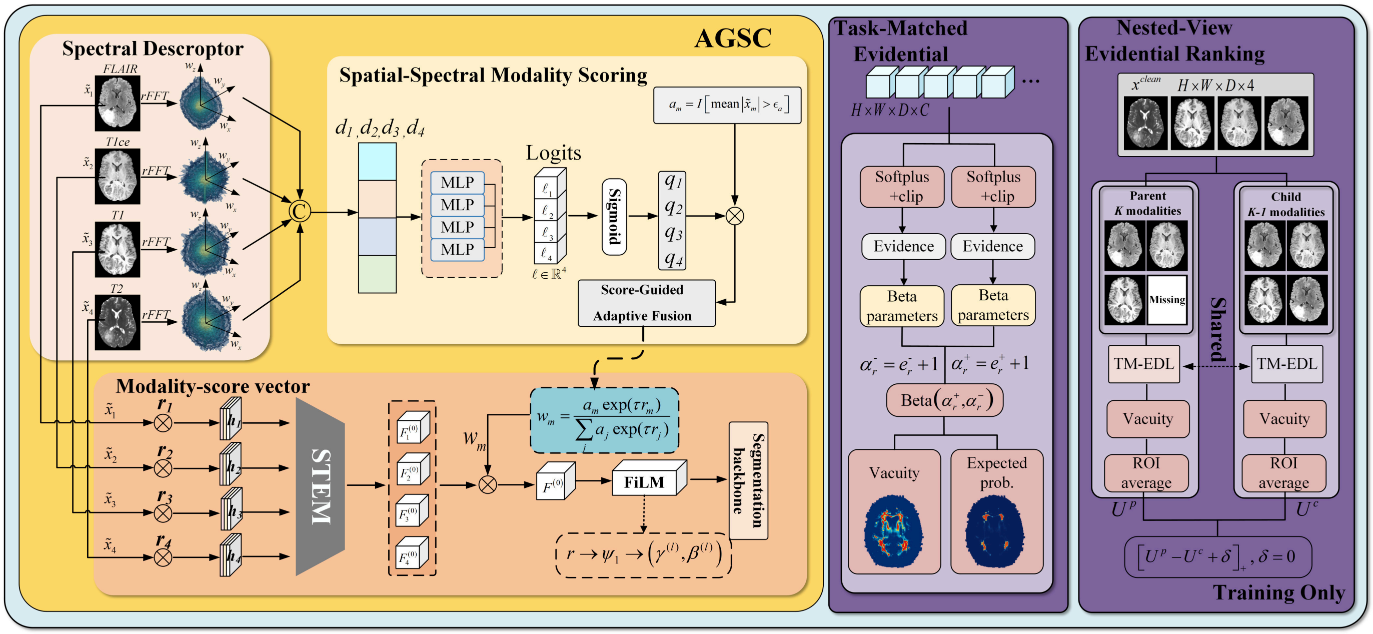

AGSC
Spectral Descroptor
FLAIR
T1ce
T1
T2
rFFT
C
Spatial-Spectral Modality Scoring
$a_m = I[\text{mean}|\tilde{x}_m| > \epsilon_a]$
d_1, d_2, d_3, d_4
Logits
MLP
MLP
MLP
MLP
ℓ_1
ℓ_2
ℓ_3
ℓ_4
$\ell \in \mathbb{R}^4$
Sigmoid
q_1
q_2
q_3
q_4
Score-Guided Adaptive Fusion
Modality-score vector
$\tilde{x}_1$
$\tilde{x}_2$
$\tilde{x}_3$
$\tilde{x}_4$
r_1
r_2
r_3
r_4
h_1
h_2
h_3
h_4
STEM
$F_1^{(0)}$
$F_2^{(0)}$
$F_3^{(0)}$
$F_4^{(0)}$
W_m
$w_m = \frac{a_m \exp(\tau r_m)}{\sum_j a_j \exp(\tau r_j)}$
$F^{(0)}$
FiLM
$r \to \psi_1 \to (\gamma^{(l)}, \beta^{(l)})$
Segmentation backbone
Task-Matched Evidential
$H \times W \times D \times C$
Softplus +clip
Softplus +clip
Evidence
Evidence
Beta parameters
Beta parameters
$\alpha_r^- = e_r^- + 1$
$\alpha_r^+ = e_r^+ + 1$
$\text{Beta}(\alpha_r^+, \alpha_r^-)$
Vacuity
Expected prob.
Nested-View Evidential Ranking
x^{clean} $H \times W \times D \times 4$
Parent K modalities
Child K-1 modalities
Missing
Shared
TM-EDL
TM-EDL
Vacuity
Vacuity
ROI average
ROI average
U^p
U^c
$[U^p - U^c + \delta]_+, \delta = 0$
Training Only

Fig. 2. Detailed illustration of AGSC, TM-EDL, and NV-Rank. Left: AGSC extracts modality-wise spatial–spectral descriptors, predicts availability-gated conditioning scores, and uses them for input attenuation, adaptive fusion, and multi-scale FiLM modulation. Middle: TM-EDL maps segmentation logits to task-compatible evidential distributions and evidential vacuity. Right: NV-Rank constructs nested parent–child modality views and penalizes violations of the prescribed vacuity ordering. NV-Rank is used only during training.

where $\mathcal{L}_{\text{seg}}$ is the deep-supervised Dice–BCE loss for BraTS region targets and Dice–CE loss for categorical tasks. $\mathcal{L}_{\text{EDL}}$ combines evidential prediction error, posterior variance, Dice supervision on evidential mean probabilities, and a target-masked KL regularizer whose weight increases linearly to 0.1 over the first 30 epochs. The indicator χ_t controls the scheduled application of NV-Rank, which is activated from epoch 5 every eight optimization steps using sampled eligible parent–child modality views. The model is trained for 300 epochs using SGD with an initial learning rate of 0.01, batch size 2, and dataset-specific patch sizes. At inference, the observed modality subset is processed through the main segmentation pathway; evidential objectives and nested-view forward passes are omitted.

IV. EXPERIMENTS

A. Experimental Setup

1) Public Datasets: We evaluate ReLiSS on BraTS2021 [1], [43], [44] for brain-tumor segmentation, ISLES2022 [45] for ischemic-lesion segmentation, and WMH2017 [46] for white-matter-hyperintensity segmentation. Table II summarizes the experimental configurations. BraTS uses WT, TC, and ET as overlapping region targets. All nonempty modality subsets are evaluated, yielding 15, seven, and three configurations for BraTS2021, ISLES2022, and WMH2017, respectively, including 22 incomplete-modality settings.

2) TSC Dataset: The private TSC cohort was retrospectively collected at Shenzhen Children's Hospital between January 2013 and January 2021 and comprised 92 pediatric patients with clinically confirmed tuberous sclerosis complex and complete FLAIR, T1-weighted, and T2-weighted MRI. MRI examinations were performed on a 3-T Siemens Skyra scanner using an 8-channel head coil. Cortical and subcortical tubers were manually delineated and cross-reviewed by multiple senior pediatric neuroradiologists. Patients with incomplete required imaging or additional neurological disorders were excluded.

Five-fold cross-validation was performed with patient-level separation, with no subject appearing in more than one fold. All seven nonempty combinations of FLAIR, T1, and T2 were evaluated, comprising six incomplete-modality configurations and one complete-modality configuration. The study was approved by the Ethics Committee of Shenzhen Children's Hospital (Approval No. 2019005), and written informed consent was obtained from the participating children and/or their legal guardians.

3) Implementation and Evaluation: We use nnU-Net preprocessing and augmentation [47], modality dropout, exact-zero filling for missing channels, and sliding-window inference. Spacing and patch sizes are listed in Table II; optimization follows Section III-F.

We compare ReLiSS with nine representative methods for missing-modality segmentation: RFNet, mmFormer, M3AE, M2FTrans, LS3M, IM-Fuse, M2SegMamba, SIUM, and UniME [4], [8]–[14], [42]. All competing methods were reproduced using their official implementations, following the recommended model configurations and training settings. Each method was evaluated under full- and missing-modality settings. Experiments were conducted on four NVIDIA RTX A6000 GPUs, each with 48 GB of memory.

We report Dice, IoU, and HD95. "Full" denotes complete inputs; "Missing" averages incomplete configurations; "Avg." also includes Full. Configurations are equally weighted, and BraTS results additionally average WT, TC, and ET. Aggregate

TABLE I
DICE SCORES (%) ON BRATS2021 ACROSS 15 NONEMPTY MODALITY CONFIGURATIONS (●: AVAILABLE; ○: MISSING).

Type	**FLAIR**	●	○	○	○	●	●	●	○	○	○	●	●	●	○	●	Avg.
	T1ce	○	●	○	○	●	○	○	●	●	○	●	●	○	●	●	
	T1	○	○	●	○	○	●	○	●	○	●	●	○	●	●	●	
	T2	○	○	○	●	○	○	●	○	●	●	○	●	●	●	●	
WT	RFNet	85.35	67.39	73.75	77.14	82.21	84.44	88.96	76.18	80.95	83.67	87.91	87.35	87.93	85.01	89.33	82.50
	M^2FTrans	86.26	72.43	77.23	78.93	83.01	83.11	89.93	76.58	79.82	81.45	86.88	88.99	87.07	86.48	89.35	83.17
	mmFormer	82.57	69.01	35.20	75.09	85.25	84.20	86.78	74.42	75.16	76.36	85.74	87.44	86.16	77.06	87.19	77.84
	M^3AE	89.93	73.23	71.31	84.26	91.42	91.39	91.50	77.64	87.08	86.49	91.90	92.45	91.76	88.00	92.68	86.74
	LS3M	88.32	75.04	74.18	84.61	88.20	87.76	88.76	78.46	87.17	87.34	88.13	88.55	88.44	87.26	90.95	85.54
	IM-Fuse	85.54	77.41	75.82	86.96	88.66	88.43	89.07	72.75	82.56	86.64	89.80	90.74	89.53	84.98	91.26	85.34
	M^2SegMamba	89.81	76.36	77.91	84.81	89.51	89.09	89.35	80.70	86.47	84.94	90.74	91.13	90.35	88.90	91.86	86.80
	SIUM	89.93	77.43	**78.13**	75.16	92.20	92.01	89.71	**81.78**	85.38	83.92	92.69	92.22	91.38	87.73	92.64	86.82
	UniME	90.14	**78.94**	77.51	84.34	92.13	90.49	91.59	81.70	88.45	**89.08**	92.01	93.22	92.66	**90.06**	92.96	88.35
	ReLiSS	**91.17**	78.76	77.96	**87.58**	**92.56**	**92.08**	**92.48**	81.66	**89.41**	88.79	**92.86**	**93.30**	**92.69**	89.89	**93.39**	**88.97**
TC	RFNet	71.79	84.77	62.08	66.42	86.08	74.76	74.90	84.20	86.41	70.25	86.56	86.41	74.93	86.35	89.31	79.01
	M^2FTrans	72.10	87.59	66.12	66.70	86.49	74.69	74.89	85.67	87.14	71.79	87.46	87.01	75.01	86.44	88.07	79.81
	mmFormer	70.62	85.79	64.71	67.17	87.77	75.30	75.38	86.18	87.48	74.09	88.53	87.82	75.50	88.61	88.26	80.25
	M^3AE	71.07	84.49	66.43	66.96	89.68	76.99	76.95	88.05	88.95	73.76	90.24	90.15	77.08	89.77	90.68	81.42
	LS3M	70.55	84.75	65.79	64.47	89.44	75.11	75.05	85.48	89.57	74.39	87.43	87.63	75.30	88.83	89.57	80.22
	IM-Fuse	71.47	87.58	64.04	69.33	87.98	76.04	76.01	87.38	87.97	**74.72**	89.71	89.89	76.15	89.29	90.89	81.23
	M^2SegMamba	72.44	86.88	68.65	72.09	89.84	76.32	76.69	89.21	89.42	73.91	90.06	89.76	77.73	90.78	91.26	82.34
	SIUM	73.18	87.65	**71.06**	64.22	89.94	**77.53**	74.87	89.33	89.20	73.62	90.36	90.31	77.62	90.07	90.57	81.97
	UniME	**73.33**	88.07	69.03	71.89	90.40	76.43	**77.35**	89.43	90.08	74.36	91.04	90.42	**78.81**	91.09	91.51	82.88
	ReLiSS	73.07	**88.44**	68.04	**72.26**	**90.83**	76.61	76.82	**90.24**	**90.62**	74.44	**91.40**	**91.19**	77.82	**91.10**	**91.55**	**82.96**
ET	RFNet	52.99	78.56	41.19	48.42	82.74	52.90	44.28	70.13	69.73	46.66	67.44	81.47	55.63	78.24	79.61	63.33
	M^2FTrans	53.05	79.12	41.08	47.80	82.96	53.20	46.90	68.19	71.88	42.28	68.42	81.43	58.76	79.91	80.54	63.70
	mmFormer	52.01	80.62	41.04	53.24	84.81	53.45	55.35	79.72	77.76	48.32	78.27	82.44	60.26	83.79	84.09	67.68
	M^3AE	52.42	79.41	48.86	54.67	83.62	51.94	52.93	82.93	83.74	49.37	84.83	84.06	59.41	84.93	85.05	69.21
	LS3M	53.15	81.88	43.63	48.72	82.73	53.36	58.95	80.19	79.73	52.71	82.23	84.07	58.06	83.83	83.52	68.45
	IM-Fuse	53.97	82.61	45.56	53.75	85.50	54.30	56.10	79.73	82.84	53.94	85.13	84.92	62.25	83.40	85.24	69.95
	M^2SegMamba	54.05	82.24	48.30	52.17	84.39	56.90	55.42	82.21	82.76	54.35	85.25	85.55	61.51	84.04	85.13	70.28
	SIUM	53.73	82.05	49.22	48.00	84.00	58.26	58.39	83.80	83.47	56.12	84.61	84.19	60.60	84.24	84.65	70.35
	UniME	**56.31**	**83.82**	**51.59**	**57.19**	85.78	**59.92**	60.02	**85.42**	85.23	**58.66**	86.15	86.00	**66.07**	86.02	**86.97**	**73.01**
	ReLiSS	55.20	83.78	49.98	55.62	**85.99**	59.46	**61.13**	85.18	**85.41**	57.80	**86.53**	**86.10**	62.20	**86.10**	86.55	72.47

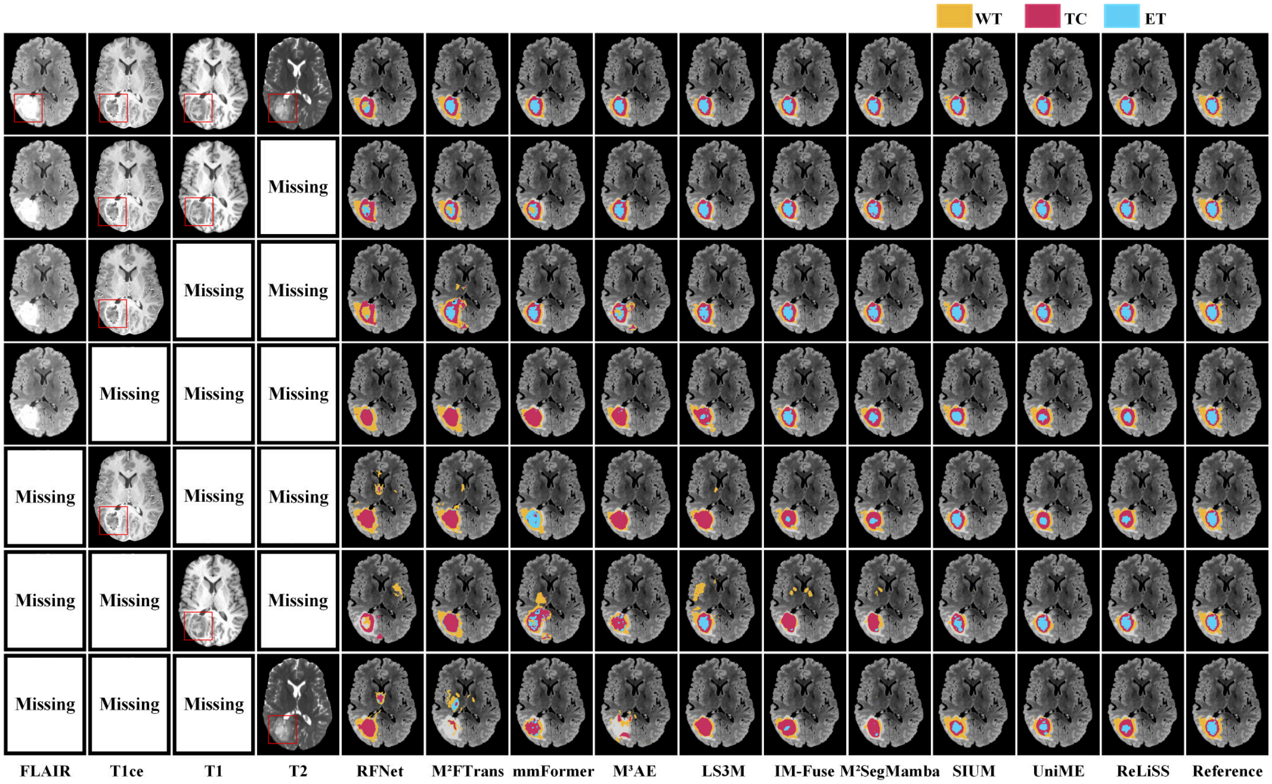


Fig. 3. Qualitative comparison on BraTS2021 under representative modality-availability configurations. Rows represent modality subsets; columns show the input modalities, competing methods, and reference segmentation.

TABLE II
SUMMARY OF DATASETS AND EXPERIMENTAL CONFIGURATIONS.

Dataset	**Cases**	**Modalities**	**Eval.**	**Spacing (mm)**	**Patch**
BraTS2021	1,251	T1/T1ce/T2/F	5-fold	1^3	128^3
ISLES2022	250	F/ADC/DWI	5-fold	2^3	80×96×80
WMH2017	60/110	F/T1	Test	3×0.98×1	48×224×192
TSC	92	F/T1/T2	5-fold	7.00×0.72×0.72	20×320×256

TABLE III
DICE SCORES (%) ON ISLES2022 ACROSS SEVEN NONEMPTY MODALITY CONFIGURATIONS (●: AVAILABLE; ○: MISSING).

FLAIR	●	○	○	●	●	○	●	**Avg.**
ADC	○	●	○	●	○	●	●	
DWI	○	○	●	○	●	●	●	
RFNet	33.82	30.10	73.32	53.30	73.44	72.96	73.88	58.69
M^{2}FTrans	37.94	34.47	72.72	58.08	74.42	73.89	74.00	60.79
mmFormer	39.90	28.21	74.16	60.51	74.99	77.14	76.00	61.56
M^{3}AE	37.25	33.57	75.94	57.74	75.84	76.85	76.81	62.00
LS3M	39.04	34.34	74.09	58.02	74.80	76.96	76.45	61.96
IM-Fuse	38.80	32.22	76.67	59.28	76.78	77.16	77.67	62.65
M^{2}SegMamba	43.11	42.32	76.92	63.24	77.79	78.35	77.34	65.58
SIUM	40.38	38.62	73.98	60.12	76.21	74.04	76.83	62.88
UniME	39.52	38.50	76.04	58.10	77.05	76.52	77.26	63.28
ReLiSS	**46.06**	**45.92**	**77.16**	**64.59**	**78.41**	**78.36**	**79.05**	**67.08**

TABLE IV
DICE (%) / HD95 (MM) ON THE OFFICIAL WMH2017 TEST SET (●: AVAILABLE; ○: MISSING).

FLAIR	●	○	●	**Avg.**
T1	○	●	●	
RFNet	72.80/6.79	56.41/12.79	73.40/6.91	67.54/8.83
M^{2}FTrans	75.34/6.37	56.73/11.33	76.07/6.05	69.38/7.92
mmFormer	76.47/6.45	60.35/11.12	76.53/6.10	71.12/7.89
M^{3}AE	75.74/6.45	59.73/11.48	76.17/6.21	70.55/8.05
LS3M	76.02/6.59	60.58/10.68	76.61/6.02	71.07/7.76
IM-Fuse	76.89/6.50	61.64/10.54	76.98/6.09	71.84/7.69
M^{2}SegMamba	76.48/6.45	62.02/10.72	76.84/6.00	71.78/7.72
SIUM	76.25/6.45	59.14/11.12	76.77/6.11	70.72/7.89
UniME	77.03/6.39	61.61/**10.48**	77.12/6.04	71.92/7.64
ReLiSS	**77.60/5.71**	**63.75**/10.98	**77.83/5.95**	**73.06/7.55**

Dice is recomputed from configuration-level scores.

B. Comparison on Public Datasets

We compare ReLiSS with state-of-the-art missing-modality segmentation methods on three public benchmarks spanning different modality counts, clinical targets, and label formulations.

1) Results on BraTS2021: Table I reports Dice across all 15 nonempty modality combinations. ReLiSS achieves average Dice scores of 88.97%, 82.96%, and 72.47% for WT, TC, and ET, respectively, with the highest observed averages for WT and TC. Across the 14 incomplete configurations, its mean Dice is 80.82%, while full-modality Dice is 90.50%. With one, two, and three modalities missing, the corresponding mean Dice values are 86.77%, 81.75%, and 73.49%.

TABLE V
DICE SCORES (%) ON TSC ACROSS SEVEN NONEMPTY MODALITY CONFIGURATIONS (●: AVAILABLE; ○: MISSING).

FLAIR	●	○	○	●	●	○	●	**Avg.**
T1	○	●	○	●	○	●	●	
T2	○	○	●	○	●	●	●	
RFNet	63.06	18.98	49.55	63.30	62.88	59.66	63.92	54.48
M^{2}FTrans	71.42	22.56	53.40	70.88	71.11	64.06	71.26	60.67
mmFormer	71.96	**26.42**	54.73	72.68	72.94	67.08	72.94	62.68
M^{3}AE	75.66	25.05	58.08	75.68	75.18	69.32	75.01	64.85
LS3M	65.75	20.93	51.15	66.12	66.05	62.47	66.46	56.99
IM-Fuse	68.61	22.26	53.37	69.07	69.33	64.26	69.44	59.48
M^{2}SegMamba	73.25	25.12	53.41	72.67	73.45	66.63	73.69	62.60
SIUM	72.94	23.54	54.71	71.80	72.98	62.35	73.51	61.69
UniME	74.39	24.43	56.44	73.98	74.21	67.86	74.29	63.65
ReLiSS	**76.38**	25.87	**58.57**	**76.90**	**76.85**	**70.64**	**77.23**	**66.06**

2) Results on ISLES2022: Table III reports all seven modality configurations. ReLiSS obtains the highest observed Dice in every configuration, with an overall average of 67.08%, a mean of 65.08% across the six incomplete settings, and 79.05% with full modalities. The gains persist across both single- and multi-modality inputs.

3) Results on WMH2017: Table IV reports results on the official WMH2017 test set. ReLiSS obtains the highest observed average Dice of 73.06% with an average HD95 of 7.55 mm, and a mean Dice of 70.68% over the two incomplete settings. With full modalities, it reaches 77.83% Dice and 5.95 mm HD95.

Figure 4 summarizes Full and Missing performance across the three public datasets. Overall, ReLiSS improves incomplete-modality robustness while preserving competitive full-modality performance across tasks with different modality counts and segmentation targets.

C. Comparison on the Private Dataset

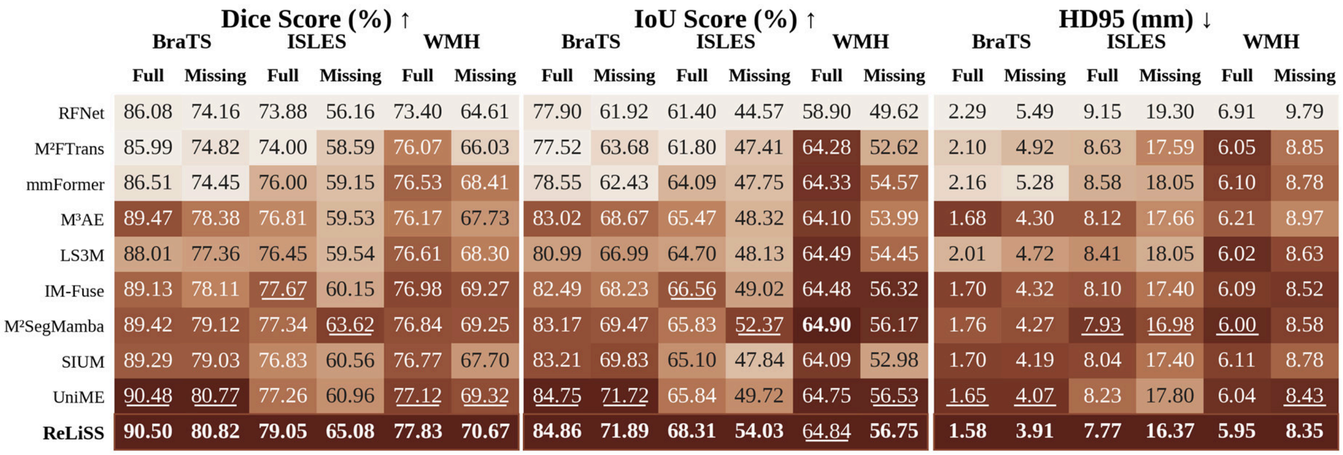
ReLiSS is further evaluated on the private TSC cohort under a clinically distinct segmentation setting.

1) Quantitative Comparison: Table V reports Dice across all seven nonempty modality combinations. ReLiSS obtains the highest observed average Dice of 66.06%, ranking first in six of seven configurations. It reaches 77.23% with full modalities and 64.20% averaged over the six incomplete configurations.

2) Full-vs-Missing Analysis: Figure 5 summarizes Full and Missing performance using Dice, IoU, and HD95. Full-modality Dice, IoU, and HD95 are 77.23%, 62.93%, and 8.46 mm, respectively; the corresponding incomplete-modality averages are 64.20%, 50.59%, and 18.97 mm. These results extend the evaluation to a distinct lesion-segmentation task.

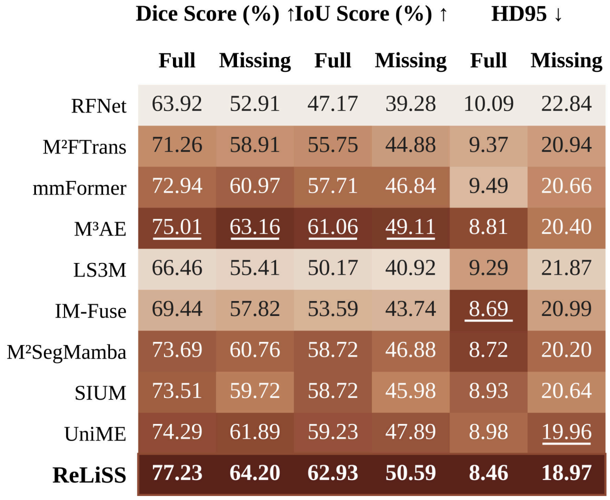
D. Ablation Studies

Four BraTS2021 ablation studies examine component contributions, modality conditioning, evidential objectives, and ranking signals. All variants are independently trained and evaluated on the same 251 validation subjects. Results are averaged over WT, TC, and ET; “Full” and “Missing-14” denote complete inputs and the macro-average over the 14



	Dice Score (%) ↑						IoU Score (%) ↑						HD95 (mm) ↓					
	BraTS		ISLES		WMH		BraTS		ISLES		WMH		BraTS		ISLES		WMH	
	Full	Missing	Full	Missing	Full	Missing	Full	Missing	Full	Missing	Full	Missing	Full	Missing	Full	Missing	Full	Missing
RFNet	86.08	74.16	73.88	56.16	73.40	64.61	77.90	61.92	61.40	44.57	58.90	49.62	2.29	5.49	9.15	19.30	6.91	9.79
M²FTrans	85.99	74.82	74.00	58.59	76.07	66.03	77.52	63.68	61.80	47.41	64.28	52.62	2.10	4.92	8.63	17.59	6.05	8.85
mmFormer	86.51	74.45	76.00	59.15	76.53	68.41	78.55	62.43	64.09	47.75	64.33	54.57	2.16	5.28	8.58	18.05	6.10	8.78
M³AE	89.47	78.38	76.81	59.53	76.17	67.73	83.02	68.67	65.47	48.32	64.10	53.99	1.68	4.30	8.12	17.66	6.21	8.97
LS3M	88.01	77.36	76.45	59.54	76.61	68.30	80.99	66.99	64.70	48.13	64.49	54.45	2.01	4.72	8.41	18.05	6.02	8.63
IM-Fuse	89.13	78.11	77.67	60.15	76.98	69.27	82.49	68.23	66.56	49.02	64.48	56.32	1.70	4.32	8.10	17.40	6.09	8.52
M²SegMamba	89.42	79.12	77.34	63.62	76.84	69.25	83.17	69.47	65.83	52.37	**64.90**	56.17	1.76	4.27	7.93	16.98	6.00	8.58
SIUM	89.29	79.03	76.83	60.56	76.77	67.70	83.21	69.83	65.10	47.84	64.09	52.98	1.70	4.19	8.04	17.40	6.11	8.78
UniME	90.48	80.77	77.26	60.96	77.12	69.32	84.75	71.72	65.84	49.72	64.75	56.53	1.65	4.07	8.23	17.80	6.04	8.43
ReLiSS	**90.50**	**80.82**	**79.05**	**65.08**	**77.83**	**70.67**	**84.86**	**71.89**	**68.31**	**54.03**	64.84	**56.75**	**1.58**	**3.91**	**7.77**	**16.37**	**5.95**	**8.35**

Fig. 4. Overall comparison of segmentation performance under full- and missing-modality settings on BraTS2021, ISLES2022, and WMH2017. Results are reported in terms of Dice (%), IoU (%), and HD95 (mm). "Full" denotes the complete-modality setting, whereas "Missing" denotes the average performance over all incomplete-modality configurations. Higher Dice and IoU and lower HD95 indicate better performance.



	Dice Score (%) ↑		IoU Score (%) ↑		HD95 ↓	
	Full	Missing	Full	Missing	Full	Missing
RFNet	63.92	52.91	47.17	39.28	10.09	22.84
M²FTrans	71.26	58.91	55.75	44.88	9.37	20.94
mmFormer	72.94	60.97	57.71	46.84	9.49	20.66
M³AE	75.01	63.16	61.06	49.11	8.81	20.40
LS3M	66.46	55.41	50.17	40.92	9.29	21.87
IM-Fuse	69.44	57.82	53.59	43.74	8.69	20.99
M²SegMamba	73.69	60.76	58.72	46.88	8.72	20.20
SIUM	73.51	59.72	58.72	45.98	8.93	20.64
UniME	74.29	61.89	59.23	47.89	8.98	19.96
ReLiSS	**77.23**	**64.20**	**62.93**	**50.59**	**8.46**	**18.97**

Fig. 5. Aggregate Full and Missing performance on TSC in terms of Dice (%), IoU (%), and HD95 (mm). Higher Dice/IoU and lower HD95 indicate better performance.

incomplete configurations, respectively. Shaded rows indicate ReLiSS.

1) Overall Contribution of the Proposed Components: Table VI evaluates cumulative additions to a backbone using zero-filled inputs, modality dropout, conventional fusion, and standard segmentation loss. AGSC raises Missing-14 Dice from 76.36% to 78.32%; FiLM lowers it to 78.01%. TM-EDL yields little change, while adding NV-Rank raises Missing-14/Full Dice to 80.05%/90.42%, exceeding the backbone by 3.69/1.68 percentage points.

2) Availability, Spectral Descriptors, and Modality Conditioning: Table VII compares SpatialOnly (without spectral cues), AvailOnly (availability-only conditioning), UniformFusion (equal weights over available modalities), and NoFiLM (without multi-scale modulation). Their Missing-14 Dice scores decrease by 0.88, 2.37, 1.10, and 1.39 percentage points relative to ReLiSS, respectively, supporting the contributions of spectral cues, adaptive fusion, and feature modulation beyond binary availability.

TABLE VI

OVERALL CONTRIBUTION OF THE PROPOSED COMPONENTS ON BRATS2021. DICE AND IOU ARE REPORTED IN %, AND HD95 IN MM. – DENOTES UNAVAILABLE RESULTS.

Method	Components				Full			Missing-14		
	AGSC	FiLM	TM-EDL	NV-Rank	Dice	IoU	HD95	Dice	IoU	HD95
Backbone	✗	✗	✗	✗	88.74	80.06	2.86	76.36	63.09	5.32
+ AGSC	✓	✗	✗	✗	89.98	82.04	2.38	78.32	65.03	4.78
+ FiLM	✓	✓	✗	✗	89.60	81.27	2.31	78.01	65.51	4.47
+ TM-EDL	✓	✓	✓	✗	89.63	81.30	2.24	77.83	65.41	4.54
ReLiSS	✓	✓	✓	✓	**90.42**	**82.94**	**1.97**	**80.05**	**66.91**	**3.96**

TABLE VII

ABLATION OF AVAILABILITY CONDITIONING, SPECTRAL CUES, ADAPTIVE FUSION, AND FILM. DICE AND IOU ARE REPORTED IN %, AND HD95 IN MM.

Method	Components			Full			Missing-14		
	Cond.	Adaptive Fusion	FiLM	Dice	IoU	HD95	Dice	IoU	HD95
SpatialOnly	Spatial	✓	✓	89.92	81.91	2.41	79.17	65.84	4.94
AvailOnly	Avail.	✓	✓	89.34	80.96	2.71	77.68	63.52	5.16
UniformFusion	Sp.+Spec.	✗	✓	89.85	81.79	2.16	78.95	65.41	4.53
NoFiLM	Sp.+Spec.	✓	✗	89.80	81.70	2.53	78.66	65.07	4.48
ReLiSS	**Sp.+Spec.**	✓	✓	**90.42**	**82.94**	**1.97**	**80.05**	**66.91**	**3.96**

3) Evidential Objectives: Table VIII compares NoEvidential, TMOnly, and NVOnly with fixed segmentation and conditioning pathways. NVOnly retains the evidential transformation but uses only the ranking objective. Either objective alone provides limited benefit; jointly, they improve Full/Missing-14 Dice from 89.60%/78.01% to 90.42%/80.05% and reduce HD95 from 2.31/4.47 to 1.97/3.96 mm, supporting their complementary roles.

4) Ranking Signal Formulation: We compare normalized entropy, $1 - (p_{(1)} - p_{(2)})$, and evidential vacuity using the same training pipeline and NV-Rank coefficient (Table IX). Vacuity achieves the highest Missing-14 Dice (80.05% versus 77.13% and 76.67%) and the lowest HD95, demonstrating its effectiveness as the ranking signal under the unified training protocol.

TABLE VIII

ABLATION OF THE TASK-MATCHED EVIDENTIAL OBJECTIVE AND NESTED-VIEW EVIDENTIAL RANKING. DICE AND IOU ARE REPORTED IN %, AND HD95 IN MM.

Method	Components		Full			Missing-14		
	TM-EDL Obj.	NV-Rank	Dice	IoU	HD95	Dice	IoU	HD95
NoEvidential	✗	✗	89.60	81.27	2.31	78.01	65.51	4.47
TMOnly	✓	✗	89.63	81.30	2.24	77.83	65.41	4.54
NVOnly	✗	✓	89.45	81.21	2.41	78.48	64.89	4.54
ReLiSS	✓	✓	**90.42**	**82.94**	**1.97**	**80.05**	**66.91**	**3.96**

TABLE IX

COMPARISON OF RANKING SIGNALS FOR NESTED-VIEW REGULARIZATION. DICE AND IOU ARE REPORTED IN %, AND HD95 IN MM.

Method	Ranking		Full			Missing-14		
	NV-Rank	Score	Dice	IoU	HD95	Dice	IoU	HD95
TMOnly	✗	–	89.63	81.30	2.24	77.83	65.41	4.54
EntropyRank	✓	Entropy	89.80	81.69	2.12	77.13	62.89	4.66
MarginRank	✓	Margin	88.83	79.90	2.71	76.67	62.57	5.14
ReLiSS	✓	**Vacuity**	**90.42**	**82.94**	**1.97**	**80.05**	**66.91**	**3.96**

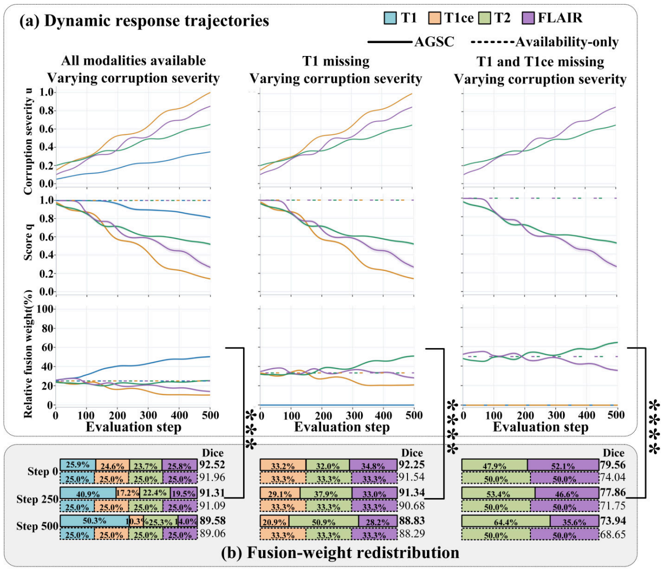


Fig. 6. Mechanistic analysis of AGSC under controlled modality degradation. (a) Dynamic response trajectories under three fixed availability settings: all modalities available, T1 missing, and T1/T1ce missing. The three rows show the prescribed corruption severity u_m, input-dependent conditioning score q_m, and relative fusion weight w_m, respectively. Solid curves denote AGSC, whereas dashed lines denote the availability-only baseline. (b) Fusion-weight compositions and corresponding Dice scores at representative evaluation steps. AGSC progressively redistributes the contributions of observed modalities as their input characteristics change, while missing modalities remain suppressed and availability-only fusion remains fixed. Statistical comparisons between AGSC and availability-only fusion use two-sided paired Wilcoxon signed-rank tests with Holm correction; *** and **** denote adjusted $p < 0.001$ and $p < 0.0001$, respectively.

E. Mechanistic and Vacuity Analysis

We further examine AGSC responses to controlled degradation at fixed modality availability and evidential-vacuity responses to progressive modality removal.

1) Input-Adaptive Conditioning under Controlled Modality Degradation: To assess conditioning beyond binary availability, we construct matched perturbation trajectories with fixed modality subsets and varying degradation of observed inputs. Figure 6(a) shows the prescribed corruption severity u_m, conditioning scores q_m, and relative fusion weights w_m. AGSC adjusts its scores and fusion weights as the inputs change, whereas availability-only fusion remains constant within each subset. Missing modalities retain zero fusion weight.

Figure 6(b) reports fusion-weight compositions and Dice scores at three representative evaluation steps. AGSC reduces the relative contribution of modalities assigned lower conditioning scores and reallocates weight to other available inputs. These responses occur under both complete and incomplete inputs, supporting adaptation to within-subset variation.

2) Evidential-Vacuity Response under Progressive Modality Removal: We next evaluate vacuity ordering under progressive modality removal. The same validation subjects are processed with nested inputs containing four, three, two, and one available modalities, enabling paired comparisons of segmentation performance and vacuity.

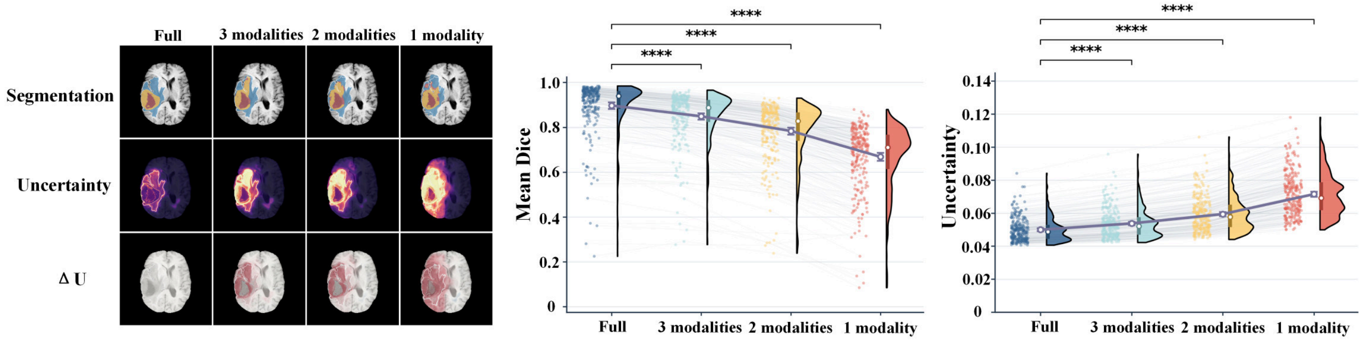
Figure 7(a) illustrates representative segmentation results, vacuity maps, and changes ΔU relative to the full-modality reference. At the cohort level [Fig. 7(b)–(c)], mean Dice decreases while mean vacuity increases along the evaluated removal sequence. This trend is consistent with the ordering encouraged by NV-Rank across nested modality views.

V. DISCUSSION

The results support combining modality-adaptive conditioning with nested-view evidential regularization. ReLiSS achieves the highest mean Dice under incomplete inputs among the evaluated methods on all four datasets. On BraTS2021, its performance is comparable to the strongest competing method, while more pronounced gains are observed on ISLES2022, WMH2017, and the private TSC cohort. The ablations provide clearer evidence for the proposed components: AGSC and FiLM increase Missing-14 Dice from 76.36% to 78.01%, and the combined evidential objectives further increase it to 80.05%. Removing spectral cues, adaptive fusion, or FiLM also reduces Dice more under incomplete than complete inputs. These findings support complementary contributions from input-adaptive conditioning and output regularization on a shared backbone.

ReLiSS achieves the highest average Dice for WT and TC, whereas its ET average is slightly below that of UniME. The ET gap is concentrated in configurations without T1ce, consistent with the role of contrast enhancement in delineating ET [43]. This gap may also partly reflect differences in representation learning, including UniME's masked pretraining followed by segmentation fine-tuning [14]. By learning cross-modal dependencies before task-specific optimization, this two-stage strategy may facilitate the use of complementary cues from the remaining modalities for ET delineation.

The evidential results support the use of vacuity as a training regularizer. TM-EDL alone provides little segmentation benefit, whereas its combination with NV-Rank improves Dice



(a)Qualitative Results **(b)Segmentation Performance** **(c)Uncertainty Response**

Fig. 7. Evidential-vacuity response under progressive modality removal on BraTS2021. (a) Representative segmentation results, vacuity maps, and vacuity changes $\boldsymbol{\Delta U}$ relative to the full-modality reference for nested inputs with four, three, two, and one available modalities. (b) Subject-level Dice and (c) evidential-vacuity distributions across modality settings. Dots denote individual subjects, thin gray lines connect paired observations, square markers and solid lines indicate cohort means with 95% bootstrap confidence intervals, and half-violin plots show the corresponding distributions. As modality information decreases, Dice declines while evidential vacuity increases, consistent with the local ordering encouraged by NV-Rank. Statistical comparisons use two-sided paired Wilcoxon signed-rank tests with Holm correction; **** denotes Holm-adjusted $\boldsymbol{p < 0.0001}$.

and HD95. Vacuity-based ranking consistently outperforms the tested entropy- and margin-based alternatives under the same training protocol, achieving higher Dice and IoU together with lower HD95. For BraTS region heads, vacuity is determined by logit magnitude; for categorical tasks, centering removes dependence on a common logit offset. Vacuity therefore serves as a task-matched confidence-related signal. The increase in vacuity under progressive modality removal is consistent with the subset ordering encouraged by NV-Rank, supporting its role as a confidence-related regularization signal.

The nested-view formulation provides a structured regularization mechanism for incomplete-modality learning by explicitly relating parent and child modality views. Unlike independent confidence supervision, NV-Rank exploits the subset relationship between inputs and encourages consistent evidential behavior as modality information changes. The improvements observed in Tables VIII and IX support the effectiveness of this subset-structured regularization when combined with TM-EDL.

Several limitations warrant further study. Missingness is simulated by withholding modalities and zero filling, and controlled degradation experiments do not establish robustness to motion, bias fields, misregistration, ghosting, or localized artifacts. Patch-wise modality scores may miss spatially heterogeneous input quality. In addition, the single-center TSC cohort provides limited evidence of generalizability across clinical settings. Future work will expand validation across centers and acquisition protocols and explore spatially adaptive conditioning under realistic imaging artifacts. [39], [48]

VI. CONCLUSION

We presented ReLiSS for multimodal MRI segmentation from arbitrary nonempty subsets of a modality set. AGSC coordinates input attenuation, adaptive fusion, and multi-scale feature conditioning, while TM-EDL and NV-Rank regularize evidential vacuity during training. Across four datasets and 28 incomplete-modality configurations, ReLiSS achieves the highest incomplete-input Dice among the evaluated methods while retaining competitive full-modality performance. These results support combining input-adaptive conditioning with nested-view regularization for segmentation with missing modalities. Robustness to acquisition artifacts and generalization to external cohorts remain to be established.

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